

Cell as Inheritance Point: The Atomic Unit Where All Paper 1 Architectural Commitments Hold Without Modification in the Coordination Knowledge Substrate Pattern

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, in operational form, the cell as Paper 2's atomic structural unit and as the inheritance point at which all Paper 1 architectural commitments hold without modification, so that downstream work can build on Paper 2's larger structures with the cell's commitments treated as Paper 1 specifies them.

Abstract

Paper 2 of the CKS theory series introduces three architectural levels — cell, aspect, Self — and develops the dynamics of how cells compose into aspects and Selves through life and evolution under unified human governance. The cell is the level at which Paper 1's specification applies directly. Paper 2 commits that all Paper 1 architectural commitments hold at the cell level without modification, making the cell the inheritance point where Series A's complete derivation chain applies operationally rather than nominally. This note formalizes that commitment: the cell is a CKS artifact with substrates, orchestration rules, and the architectural commitments Paper 1 defends; it handles a specific informational task with scope bounded by purpose; it is the modular Lego-piece that Paper 2's larger structures compose from. The note enumerates the Paper 1 commitments inherited at cell level (A1.01–A1.16), articulates what makes the modular-Lego-piece framing architectural rather than metaphorical, identifies the operational implications, and bounds the inheritance to cell scope.

1. Why the cell as inheritance point needs to be formalized

Paper 2 develops a three-level structural architecture — cell, aspect, Self — and commits that Paper 1's architectural commitments hold at every level. The commitment at the cell level is structurally distinct from the commitments at higher levels: the cell is Paper 1's atomic unit. Paper 2 does not redefine what a cell is; it inherits the cell from Paper 1 and develops what cells compose into. Naming the cell explicitly as the inheritance point is what makes Paper 1's complete specification operationally available to Paper 2 deployments. Without the explicit naming, a reader of Paper 2 might infer that Paper 2 introduces a new kind of cell, distinct from Paper 1's, with its own architectural commitments to defend. The opposite is true: the cell in Paper 2 is exactly the cell from Paper 1, with all sixteen foundational architectural commitments enumerated in Series A (A1.01 through A1.16) holding at cell level without modification.

This is the third Phase B1 foundational note. The first establishes the unifying architectural move (instinct/reasoning separation as independently-evolving layers); the second establishes the structural framework (three levels with relational role membership); this note establishes the cell as the inheritance point where Paper 1 applies. The next two Phase B1 notes establish aspect and Self with their level-distinct treatments. The recursive-commitments-across-all-levels treatment is formalized as standalone in B1.20.

2. The architectural commitment

In the CKS Paper 2 architecture, the **cell** is the atomic structural unit inherited from Paper 1. The commitment has four parts.

The cell is the atomic unit from Paper 1. A cell is a CKS artifact comprising substrates (persistent, structured, human-governed representations of coordination knowledge), orchestration rules (human-authored rules determining cell-level behavior), and the LLM-mediated execution that operates over them under those rules. This is exactly the cell of Paper 1 §2.1; Paper 2 does not redefine the term.

All Paper 1 architectural commitments hold at the cell level without modification. The sixteen foundational commitments enumerated in Series A — substrate-cell boundary, human-governance, conflict-as-first-class, AI-as-substrate-mediator, tool-agnosticism, linear-cost scaling, path retraceability, substrate-as-source-of-truth, KO/OIDA inheritance, determinism contract, non-specialist governance, labor allocation, composition requirements, three adjacencies, orchestration layer distinctions, hybrid composition — hold at cell level identically to the way Paper 1 defends them. There is no modification, weakening, or refinement at cell scope.

The cell handles a specific informational task. A cell's scope is bounded by the purpose for which it exists. Multiple cells exist within an aspect, and within a Self, because each handles its specific work. The cell is not a Self; it is not an aspect; it is the unit that performs a single bounded informational task under orchestration rules.

The cell is the modular Lego-piece. Cells are the modular units that Paper 2's larger structures (aspects and Selves) compose from. Paper 1 specifies the piece; Paper 2 specifies what those pieces build into and how those builds develop and change over time. Modularity at the cell level is architectural commitment from the start, not an evolved property the architecture has to wait for.

The commitment is *operational*: the inheritance is not a nominal claim that Paper 2 cells are "like" Paper 1 cells, but the architectural claim that every operational property Paper 1 establishes — every test, every anti-pattern, every composition constraint — applies to cells in Paper 2 deployments without need for re-derivation.

3. What makes the cell-as-inheritance-point architecturally distinctive

Three properties distinguish Paper 2's cell-level commitment from a looser claim of architectural similarity.

Paper 2 builds on Paper 1, not parallel to it. Paper 2's three-level structure stacks on top of Paper 1's atomic-unit specification. The cell is the bottom of Paper 2's stack and the entirety of Paper 1's structural object. Aspects and Selves are larger structures composed of cells; they are not redefinitions of cells, and they do not replace Paper 1's commitments at cell scope.

Series A applies to cells in Paper 2 deployments. The full derivation chain of Series A — Phase A1 foundational commitments, Phase A2 operational variants, Phase A3 anti-patterns, Phase A4 composition pairs, Phase A5 operational tests, Phase A6 boundary cases — applies to cells in Paper 2 systems. A deployment that instantiates Paper 2's architecture inherits Series A's complete coverage of cell-level architectural questions for free; the work of Series A is reusable rather than redoable.

The inheritance is operational, not just theoretical. Every Series A operational test verifies a Paper 1 architectural property at cell scope; running these tests at cell level in Paper 2 deployments verifies the inheritance directly. The substrate-cell boundary holds at cell level; human-governance holds at cell level; conflict preservation holds at cell level; and so on through all sixteen commitments. Paper 2 systems pass Paper 1's operational tests at cell level by construction, because the cell is Paper 1's specification.

These three properties together distinguish "Paper 2 cells *are* Paper 1 cells" from a weaker claim of "Paper 2 cells resemble Paper 1 cells." The former is what Paper 2 commits to; the latter would require independent defense.

4. The modular Lego-piece framing

Paper 2 frames the cell as a modular Lego-piece. The framing is architectural rather than metaphorical and carries three points of substantive content.

Paper 1 specifies the piece. What a cell is — its substrates, its orchestration rules, the LLM-mediated execution under those rules, the architectural commitments enumerated in Series A — is specified by Paper 1 and inherited at cell level by Paper 2 without modification. Paper 2 does not respecify the

piece.

Paper 2 specifies builds and dynamics. What pieces build into (aspects, Selves), how those builds compose under unified governance, and how those builds develop and change over time (lifecycle primitives of birth, mating, death; the three evolution mechanisms of instinct, DNA, action-feedback; multi-level evolution; horizontal and vertical evolution) is the work of Paper 2. Paper 2 starts where Paper 1 ends — at the modular cell — and develops the structures and dynamics modularity makes available.

Modularity is architectural commitment from the start, not evolved property. Biology had to evolve modular architectures over deep time because modularity supports evolvability. Modular body plans — segmentation, organ systems, distinct tissue types with relational roles — emerged through long evolutionary trajectories before they could function as scaffolds for further evolution. CKS does not face this constraint. CKS is modular by architectural commitment from the cell up; the modular Lego-piece exists at instantiation. Every property the lifecycle and evolution mechanisms depend on — addressability of cells, governable origination, governable retirement, the ability to combine cells under orchestration — is available as a property of the cell from instantiation.

The third point is itself a foundational commitment with substantive content beyond the present note; it will be formalized as standalone in B1.08. The framing is introduced here because it is what makes the cell-as-Lego-piece commitment architectural rather than metaphorical, and what makes Paper 2's lifecycle and evolution machinery coherently definable.

5. The inherited commitments at cell level

The sixteen Paper 1 foundational architectural commitments inherited at cell level, with their operational meaning each, are as follows.

A1.01 — Human-governed (authority, not labor). Humans retain rights to inspect, modify, and override any cell-level substrate content and any orchestration rule governing the cell, at all times.

A1.02 — Substrate-cell boundary. The cell preserves the architectural separation between substrate (state) and cell (rule-governed execution); substrate-relevant state lives in substrate, not in the cell's execution context.

A1.03 — Conflict as first-class object. Contradictions and unresolved ambiguities at cell level are preserved as first-class substrate state; resolution decisions are made at the cell layer under orchestration rules and recorded back to substrate.

A1.04 — AI as substrate mediator. Within the cell, the LLM operates as substrate mediator under orchestration rules — reading from and writing to substrate, with authority bounded by what humans govern.

A1.05 — Tool-agnosticism. The cell's substrate may be hosted on any environment satisfying the three minimal requirements (persistent state, human read/write access, LLM access).

A1.06 — Linear-cost scaling. Cell execution cost is proportional to the cell's bounded scope, not to total substrate size; the cost contract from Paper 1 §6 holds at cell level.

A1.07 — Path retraceability. Every cell-level write carries the six-field provenance metadata; cell operations are retraceable through substrate addressability.

A1.08 — Substrate as source of truth. The substrate is authoritative for *what is the case* about the work the cell coordinates; cell-internal state, LLM context, and any other ephemeral channel are not.

A1.09 — KO/OIDA inheritance. The two-axis extension — governance plus multi-human — applies at cell scope; the cell inherits the OIDA observe-orient-decide-act loop and the Knowledge Object representational discipline.

A1.10 — Determinism contract. Substrate state plus rules plus inputs determine cell-level outputs in the senses Paper 1 §4 and §6 specify, with allowed non-determinism scoped to LLM-mediated reasoning the substrate captures.

A1.11 — Non-specialist governance. The cell's governance — inspect, modify, override — is exercisable in commodity tools by non-specialists, decoupled from authorship expertise.

A1.12 — Labor allocation. The work of authoring and maintaining cell-level content is allocable across three modes (direct human, LLM-under-rule, stable-cell); the architecture supports all three and requires none in particular.

A1.13 — Composition requirements. When cells compose, the constraints from Paper 1 (no shared mutable state outside substrate, communication through substrate, governance scope per arrangement, and the others enumerated in A1.13) apply to compositions involving the cell.

A1.14 — Three adjacencies. The cell is distinct from RAG, parametric memory, and external structured memory in the architectural senses Paper 1 §1 enumerates; cell-level work is none of these.

A1.15 — Orchestration layer distinctions. The cell is distinct from a workflow engine step, an agent framework node, and a control-plane operation in the architectural senses Paper 1 §8 enumerates.

A1.16 — Hybrid systems composition. When the cell composes with adjacent AI components (vector stores, classifier models, training pipelines, and others), the three composition patterns from Paper 1 (input, derived view, separate concern) apply; the cell does not absorb adjacent components into its boundary.

The enumeration above lists the foundational sixteen, each of which the cell in Paper 2 inherits without modification. Series A's full derivation chain extends through operational variants, anti-patterns, composition pairs, operational tests, and boundary cases beyond these sixteen; those further derivations apply at cell level in Paper 2 systems by the same inheritance.

6. Operational implications and limits

Three operational consequences and three limits follow directly.

Operational implication 1 — Series A applies to cells in Paper 2 systems. Deployments using Paper 2's architecture do not need to re-derive any Paper 1 architectural commitment for cells; Series A's notes apply directly. A deployment auditor verifying that a Paper 2 system preserves Paper 1's commitments at cell level can refer to Series A as the standing derivation chain.

Operational implication 2 — Series A's operational tests verify cell-level inheritance. The Series A operational tests provide concrete verification procedures for Paper 1 commitments at cell scope; running these tests at cell level in Paper 2 deployments verifies the inheritance is preserved. Series B will introduce additional operational tests for aspect and Self levels with their level-distinct scope.

Operational implication 3 — Cells are the operational anchor for verification. Because the cell is the inheritance point and Series A applies directly, deployment verification of Paper 1 properties happens at cell level first. Aspects and Selves verify their own level-distinct properties as those built on cells; verifying cell-level properties is logically prior because higher-level properties inherit from and depend on what cells preserve.

Limit 1 — The unmodified-inheritance property applies AT cell level. Paper 1 commitments hold at cell level without modification. They also hold at aspect and Self levels per Paper 2's recursive-commitments principle, but with level-distinct scope. The unmodified-inheritance property is specific to the cell level and should not be read as collapsing aspects and Selves into "just cells."

Limit 2 — Aspects and Selves have level-distinct treatments. Aspect-level commitments operate at aspect scope (relational role membership of cells, purpose-defined coordination across cells, aspect-level governance); Self-level commitments operate at Self scope (integrated whole holding multiple aspects, Self-level lifecycle, multi-level evolution under unified governance). These are developed in B1.04 and B1.05.

Limit 3 — The cell is not the only structural level. Paper 2 develops three levels precisely because the cell alone does not cover the architectural ground Paper 2 commits to. Aspects coordinate cells for purposes; Selves integrate aspects into wholes. The cell-as-inheritance-point commitment is the foundation, not the entirety, of Paper 2's structural architecture.

7. Operational test

A deployment instantiates the cell-as-inheritance-point commitment if and only if all of the following are true at all times during a cell's existence:

1. Each cell satisfies the operational tests for the sixteen Paper 1 foundational architectural commitments (A1.01–A1.16) at cell scope, with no commitment weakened, refined, or replaced relative to Paper 1's specification.
2. Each cell handles a specific informational task with scope bounded by the cell's purpose, distinct from other cells' tasks within the same aspect or Self.

3. Each cell is composable into larger structures (aspects, Selves) under the composition requirements (A1.13) and orchestration layer distinctions (A1.15) that Paper 1 defends, without absorbing the larger structures' architectural distinctness into the cell.
4. The substrate-cell boundary (A1.02) holds at the cell's boundary identically to Paper 1's specification: substrate carries state, cell carries rule-governed execution, cell-to-cell communication runs through substrate.
5. Each cell exists as a modular unit available to Paper 2's lifecycle operations and evolution mechanisms at instantiation, not as a property the architecture has to wait for or evolve into.

A system that fails any of (1)–(5) may instantiate one or more Paper 2 properties at higher structural levels, but does not preserve the cell-as-inheritance-point commitment, and Paper 1's architectural defense does not transfer to its cells without independent verification.

8. Why naming this commitment as standalone matters

The cell-as-inheritance-point property is what makes Series A directly relevant to Series B deployments. Without the property, Paper 2 would need to re-derive every Paper 1 commitment for its cells; Series A's coverage would not transfer without independent argument; deployments would carry the burden of verifying every cell-level architectural property from scratch. With the property, Series A applies as-is to cells in Paper 2 systems, and Series B builds on top with level-distinct work for aspects and Selves.

The modular Lego-piece framing is what makes Paper 2's lifecycle and evolution mechanisms coherently definable. Birth, mating, and death of cells; the three evolution mechanisms operating on cells; multi-level evolution stacking on cell-level evolution — all depend on the cell being a modular unit at instantiation. Without modularity from architectural commitment, life and evolution operations are not coherently definable; with it, they are.

Subsequent Phase B1 notes elaborate the structural framework: B1.04 develops aspect as coordination arrangement of cells for purpose; B1.05 develops Self as integrated whole holding multiple aspects. B1.08 will formalize modularity-from-architectural-commitment as standalone, picking up the framing introduced here. B1.20 will formalize recursive-Paper-1-commitments across all levels, generalizing the cell-level inheritance treated here to aspect and Self levels with level-distinct scope.

Subsequent work that adopts Paper 2's architecture, extends it, composes it with adjacent patterns, or argues against it should treat cells as Paper 1's atomic unit with the architectural commitments Paper 1 defends, holding at cell level without modification, and as the modular Lego-piece Paper 2's larger structures compose from. Subsequent work that uses *cell* in a different sense at the foundational level is using a different concept, and the difference should be named.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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